MAPLE: Modular Attention for Interpretable and Prosocial Multi-Agent Reinforcement Learning

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Keywords: Multi-agent Reinforcement Learning, Modular Neural Networks, Parameter-efficient Fine-tuning, Interpretable AI

Summary

We propose a novel approach to enhance interpretability and performance in multi-agent reinforcement learning (MARL) through modular architecture and representation learning. We introduce MAPLE (Modular Attention for Prosocial Learning), a MARL architecture built on Independent Proximal Policy Optimization (IPPO) with brain-inspired modular processing that mirrors the functional specialization observed in human neural systems. We incorporate a pre-trained slot attention model to learn compositional, object-centric representations, along with modular recurrent networks which interact through an attention bottleneck, consistently outperforming end-to-end RL across different environments. Our architecture incorporates three key mechanisms to address challenges in complex social scenarios: (1) learning orthogonal latent representations and directing them through modular recurrent neural networks, allowing different modules to specialize in processing distinct environmental features, (2) enabling generalization via compositional consistency loss for slot attention modules, and (3) using different fine-tuning approaches such as LoRA and progressively unfreezing parts of the slot attention module during RL training, allowing pretrained representations to adapt while maintaining their specialized structure. Moreover, we integrate modular Recurrent Independent Mechanisms (RIMs) in agents' value networks, encouraging sparsity and fast adaptation to changing environments, with modules dynamically activating based on relevant environmental contexts. This unique combination of attention mechanisms and modular processing results in specialized neural components that focus on different environmental aspects, thus improving coordination, generalization, and interpretability. We evaluate MAPLE in DeepMind's Melting Pot suite across three environments: mixed strategies (Prisoner's Dilemma), social good (Allelopathic Harvest), and resource management (Territory Room). Our results demonstrate that MAPLE not only enhances performance but also reveals emergent social capabilities across environments while providing deeper insights into the learned representations driving agent behaviors.

Contribution(s)

1. This paper presents MAPLE (Modular Attention for Prosocial Learning), a novel architecture for multi-agent reinforcement learning that combines pre-trained slot attention with recurrent independent mechanisms to enhance both interpretability and performance.
   Context: Prior work in MARL has often sacrificed interpretability for performance, using end-to-end neural networks that learn entangled representations.
2. We introduce a modular object-centric representation learning approach with specialized fine-tuning strategies that maintains the benefits of pre-trained representations while allowing adaptation to dynamic multi-agent environments.
   Context: Existing approaches typically use either fixed pre-trained representations or fully adaptable models, without effectively combining the benefits of both.
3. We demonstrate through information-theoretic evaluations that our modular approach enables specialized neural components to naturally develop representations of other agents' behaviors, providing quantitative evidence for emergent social capabilities without explicit modeling.
   Context: Previous social MARL approaches have often required explicit modeling of other agents rather than allowing social capabilities to emerge naturally from architecture design.

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Abstract

We propose a novel approach to enhance interpretability and performance in multi-agent reinforcement learning (MARL) through modular architecture and representation learning. We introduce MAPLE (Modular Attention for Prosocial Learning), a MARL architecture built on Independent Proximal Policy Optimization (IPPO) with brain-inspired modular processing that mirrors the functional specialization observed in human neural systems. We incorporate a pre-trained slot attention model to learn compositional, object-centric representations, along with modular recurrent networks which interact through an attention bottleneck, consistently outperforming end-to-end RL across different environments. Our architecture incorporates three key mechanisms to address challenges in complex social scenarios: (1) learning orthogonal latent representations and directing them through modular recurrent neural networks, allowing different modules to specialize in processing distinct environmental features, (2) enabling generalization via compositional consistency loss for slot attention modules, and (3) using different fine-tuning approaches such as LoRA and progressively unfreezing parts of the slot attention module during RL training, allowing pretrained representations to adapt while maintaining their specialized structure. Moreover, we integrate modular Recurrent Independent Mechanisms (RIMs) in agents' value networks, encouraging sparsity and fast adaptation to changing environments, with modules dynamically activating based on relevant environmental contexts. This unique combination of attention mechanisms and modular processing results in specialized neural components that focus on different environmental aspects, thus improving coordination, generalization, and interpretability. We evaluate MAPLE in DeepMind's Melting Pot suite across three environments: mixed strategies (Prisoner's Dilemma), social good (Allelopathic Harvest), and resource management (Territory Room). Our results demonstrate that MAPLE not only enhances performance but also reveals emergent social capabilities across environments while providing deeper insights into the learned representations driving agent behaviors.

1 Introduction

Multi-Agent Reinforcement Learning (MARL) requires handling the complexities of sequential decision-making in environments populated by multiple agents. The primary difficulty originates from the non-stationarity of these environments, as agents continually update their policy parameters (Papoudakis et al., 2019). Each agent must adjust its behavior not only in response to its physical environment but also to other agents' actions. This dynamic makes MARL highly complex, often resulting in interactions and decisions that are difficult to interpret or generalize to new settings. In real-world applications, such as autonomous driving or multi-robot systems, understanding and explaining agents' decision-making is critical for ensuring safety, troubleshooting, and enhancing coordination.

Recent advances in deep MARL have produced impressive results in complex domains (Baker et al., 2022; Samvelyan et al., 2019), but often at the cost of interpretability, as end-to-end neural networks typically learn entangled representations where the reasoning behind specific decisions remains opaque. This interpretability gap motivates our research into brain-inspired modular architectures that can maintain performance while offering insights into agent behavior. The functional specialization of the human brain, where distinct neural regions process different aspects of perception and cognition, provides a compelling exemplar for artificial intelligence systems (Hassabis et al., 2017; Kriegeskortea & Douglas, 2018). Neural evidence suggests that the brain organizes information processing into specialized modules that coordinate through attention mechanisms (Bullmore & Sporns, 2012). By mimicking this modular organization, we can create AI systems that not only perform effectively but also offer more transparent decision-making processes. Cognitive science research highlights the fundamental role of attention in shaping understanding.

> "Our sense of self is deeply tied to where we direct our attention. By consciously choosing where to focus, we shape our reality, our thoughts, and ultimately, our identity."
>
> *Bernard Baars*

Inspired by this cognitive science perspective, we propose a modular approach to MARL that leverages object-centric representations through Slot Attention (Locatello et al., 2020; Seitzer et al., 2023) and Recurrent Independent Mechanisms (RIMs) (Goyal et al., 2019; 2022). Slot Attention decomposes environments by assigning input data to specialized slots representing distinct entities, while RIMs provide specialized processing modules activated selectively based on input relevance. Together, these mechanisms enable agents to develop specialized representations of different environmental aspects, promoting both interpretability and adaptability to complex, dynamic scenarios.

Our model, **MAPLE** integrates pre-trained slot attention with a bilevel optimization strategy and modular recurrent mechanisms in policy networks (Jia et al., 2023), to promote both interpretability and performance by disentangling complex environmental signals and encouraging robust and adaptable representations for decision-making. During the pre-training process, the outer objective focuses on reconstruction, while the inner objective functions as a soft-clustering mechanism. We further enforce disengagement in Query-Slot Attention (QSA) with orthogonal projection loss (OPL)(Ranasinghe et al., 2021) to ensure each slot learns unique features. Additionally, we apply a compositional consistency loss between the encoder and the decoder to promote compositional generalization of object-centric representations (Wiedemer et al., 2024).

Each slot's representation is processed by separate recurrent modules with LSTM/GRU layers to learn latent feature histories and cause-effect relationships, enabling reliable event prediction and declarative knowledge acquisition (Goyal et al., 2021). In our policy network, we replace the RIM architecture's encoder and input attention bottleneck with a slot module that passes latent representations directly to recurrent modules. While the original RIM selectively activates only the k most relevant recurrent units, slot attention distributes input features across all slots without this top-k mechanism, making the model more inclusive of information. This improves interpretability and feature integration by ensuring all recurrent modules are updated, promoting specialization. Our policy's modular attention design resembles the Relational Memory Core (RMC) (Santoro et al., 2018), where memory slots interact with each other through multi-head attention without sparse gating, allowing for more comprehensive information exchange.

Our architecture employs a hybrid approach: while the policy network uses slot attention for comprehensive feature integration, the value network retains the RIM architecture with its original attention bottleneck for computational efficiency. The slot attention mechanism is pretrained on random crops from bird's-eye views of the mini-grid world matching agent observation size. During MARL training, each agent begins with this pretrained foundation and fine-tunes its own representation through environmental interaction. We implement Low-Rank Adaptation (LoRA) (Hu et al., 2022) to reduce computational costs by fine-tuning only selected layers, enabling efficient decentralized training and representation specialization.

Our key contributions include: (1) a novel brain-inspired modular architecture that combines slot attention with recurrent mechanisms to enhance both interpretability and performance in MARL; (2) specialized fine-tuning strategies that maintain the benefits of pretrained representations while allowing adaptation to multi-agent dynamics; and (3) empirical evidence that modular approaches can promote more interpretable and prosocial behaviors in complex environments.

We evaluate MAPLE in diverse social settings using DeepMind's Melting Pot 2.0 environment (Agapiou et al., 2023), testing its ability to generalize across varied social contexts with novel partners (Leibo et al., 2021). This approach aligns with our goal of investigating multi-agent dynamics in complex, socially rich environments while maintaining interpretability. We analyze information integration in the modular recurrent layers using information-theoretical metrics (Jaques et al., 2019), providing insights into how modular systems coordinate to produce effective behaviors.

In the following sections, we present related work, detail our MAPLE methodology, describe our experimental design, present results, and discuss implications of our findings.

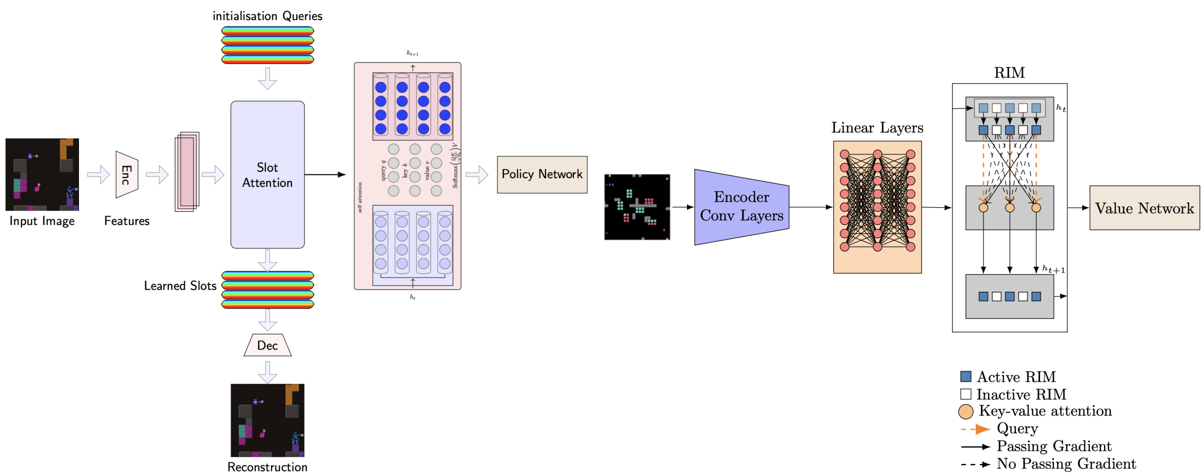

Figure 1: Architectural illustrations of our model components. **Left:** The actor network processes each agent's observation through a slot attention mechanism, mapping the slot representations to separate recurrent modules to obtain the updated policy head. **Right:** The value network consists of an encoder with the latent layer passing to the RIM module before entering the value head.

2 Related Work

Prior work in object-centric MARL has explored various representational approaches. Shang et al. introduced agent-centric attention mechanisms for relational reasoning that enable complex cooperation strategies (Shang et al., 2021), while Nath et al. applied object-centric methods for single-agent RL by extracting generalized value functions from pixel inputs (Nath et al., 2023). Liu et al. achieved scalability through "semantic tracklets" as disentangled representations (Liu et al., 2021). These approaches align with broader efforts to develop inductive biases for higher-level cognition in deep learning systems (Goyal & Bengio, 2023). Our MAPLE framework uniquely combines pre-trained slot attention with modular recurrent networks to ensure specialized processing of environmental features.

Wei et al. proposed Ranked Policy Memory (RPM), a self-play framework maintaining a repository of past policies (Qiu et al., 2023), whereas MAPLE improves representational learning through modular coordination without external policy memory. Oldenburg et al. promoted cooperative behavior through Bayesian norm learning (Oldenburg & Zhi-Xuan, 2024), while Vinitsky et al. demonstrated the efficacy of public sanctioning (Vinitsky et al., 2023).

Recent studies have addressed plasticity loss in deep RL (Sokar et al., 2023; Elsayed et al., 2024; Arthur Juliani, 2024). MAPLE extends these ideas by tracking dormant neurons within fine-tuned slot attention modules. Unlike prior approaches focusing on either architectural innovations or social dynamics separately, MAPLE integrates brain-inspired modularity, object-centric representations, and efficient fine-tuning strategies to simultaneously enhance interpretability, performance, and prosocial behaviors in complex multi-agent environments.

3 Preliminaries

Our multi-agent approach builds on Independent Proximal Policy Optimization (IPPO) (Witt et al., 2020; Yu et al., 2022), extending PPO (Schulman et al., 2017) to multi-agent settings. In this framework, each agent trains independently with dedicated actor and critic networks, enabling decentralized learning while sharing an environment. Notably, agents do not share any network structure and optimize solely for their individual rewards, maintaining complete policy independence. The policy network processes local observations (O_t^l), while the value network utilizes global observations (O_t^g) to improve credit assignment (Lowe et al., 2017).

MAPLE explores two configurations within this framework. In the first, a conventional encoder—comprising convolutional layers with positional encoding—extracts features that feed into a modular memory component based on the Recurrent Independent Mechanisms (RIM) architecture (Goyal et al., 2019) before the policy head. The second configuration replaces this encoder with a pretrained bi-level slot attention module (Jia et al., 2023), where slot initializations are optimized during pretraining using reconstruction loss and regularized by orthogonal projection and compositional consistency losses. This approach more closely resembles the brain's object-centric visual processing, where different neural populations selectively respond to distinct features and objects in the visual field (Roelfsema & R., 2011).

We employ slot attention specifically to enforce orthogonal object-centric features in each recurrent module, substantially enhancing interpretability. By directing each slot to attend to distinct environmental entities, we create a natural alignment between slot representations and meaningful components, allowing us to trace how specific objects influence policy decisions through dedicated processing pathways.

During fine-tuning in the second configuration, we remove the RIM's input cross-attention and instead pass each slot's content directly to independent recurrent modules while preserving the RIM's communication attention. The resulting RL objective is augmented with regularizers:

$$\mathcal{L} = \mathcal{L}_{\text{IPPO}} + \lambda_{\text{orth}}\mathcal{L}_{\text{OPL}} + \lambda_{\text{cons}}\mathcal{L}_{\text{CCL}}, \tag{1}$$

where $\mathcal{L}_{\text{IPPO}}$ is the standard IPPO loss, $\mathcal{L}_{\text{OPL}}$ promotes orthogonality among slot representations, and $\mathcal{L}_{\text{CCL}}$ enforces consistency between input encoding and its reconstruction. The value network retains a convolutional encoder followed by the original RIM architecture.

For computational efficiency with quality representation, we selectively fine-tune only the layers closest to the slots in the pretrained slot attention module. We employ a dual learning rate approach: a different rate for the slot attention module versus the rest of the policy network, enabling careful adaptation of object-centric representations while allowing policy components to learn at their regular pace. We apply Generalized Advantage Estimation (Schulman et al., 2016) to stabilize value updates and improve sample efficiency by taking a weighted average of temporal difference residuals over multiple time steps.

3.1 Slot Attention

Our approach leverages Bi-level Optimized Query Slot Attention (BO-QSA) to encode object-centric representations from observations. Rather than relying solely on a discrete VAE for tokenization, BO-QSA initializes slots with learnable queries that are iteratively refined via attention mechanisms and recurrent updates. During pretraining, the module is optimized with reconstruction loss plus orthogonal projection and compositional consistency regularizers to encourage distinct slot representations (Jia et al., 2023). This design mitigates random initialization drawbacks and enables effective clustering of features into meaningful object concepts.

During RL fine-tuning, we replicate the pretrained slot attention module for each agent and selectively update high-level layers—projection layers for queries, keys, values, and slot layer—using LoRA or partial unfreezing, while preserving low-level features. This selective adaptation mirrors biological visual processing, where low-level features remain relatively fixed while higher-level processing adapts to specific tasks (Yamins & DiCarlo, 2016). This modularity facilitates distributed, specialized processing critical for handling complex, dynamic environments (Mittal et al., 2022).

3.1.1 Orthogonal Projection Loss and Consistency Loss

We encourage inter-class separation and intra-class compactness through an Orthogonal Projection Loss (OPL) (Ranasinghe et al., 2021). This regularizer enforces orthogonality on learned slot features, ensuring each slot captures distinct representations by maximizing intra-slot similarity while minimizing inter-slot similarity. The mean dot product between feature vectors of the same slot is:

$$\mu_{\text{pos}} = \frac{\sum_{i,j} M_{\text{pos}}(i,j)\langle f_i, f_j\rangle}{\sum_{i,j} M_{\text{pos}}(i,j) + \epsilon}; \quad \langle f_i, f_j\rangle = \frac{f_i \cdot f_j}{\|f_i\|\|f_j\|} \tag{2}$$

where f_i and f_j are feature vectors, and M_{pos} equals one when vectors belong to the same slot. Similarly, the mean dot product between different slots is:

$$\mu_{\text{neg}} = \frac{\sum_{i,j} M_{\text{neg}}(i,j)\langle f_i, f_j\rangle}{\sum_{i,j} M_{\text{neg}}(i,j) + \epsilon} \tag{3}$$

with M_{neg} equal to one for features from different slots. The final loss combines these:

$$L_{\text{OPL}} = (1 - \mu_{\text{pos}}) + \mu_{\text{neg}} \tag{4}$$

To further improve slot learning, we employ a compositional consistency loss that promotes learning invariant object-centric representations that generalize compositionally (Wiedemer et al., 2024). This regularizer ensures the encoder ϕ can correctly invert the decoder ψ when given reconstructions of novel combinations of learned object representations z:

$$\mathcal{L}_{\text{cons}}(\phi, \psi, z') = \mathbb{E}_{z'}\left[\|\phi(\psi(z')) - z'\|_2^2\right] \tag{5}$$

This loss enforces a crucial property for MARL: when slots representing different objects are recombined in new configurations (as happens naturally in multi-agent environments), the encoder should still map them correctly to their original latent representations. By minimizing this consistency error during both pretraining and fine-tuning, we encourage each recurrent module to learn specialized, robust object-centric features that maintain semantic meaning regardless of context, enhancing transfer when agents encounter novel configurations. After obtaining slot representations, we apply layer normalization before passing them to recurrent modules, ensuring consistent scale across different slot features.

3.2 Recurrent Module

We incorporate independent recurrent modules to enable agents to track temporal changes in disentangled objects. The value network uses Recurrent Independent Mechanisms (RIMs) with independent units that interact sparsely through an attention bottleneck.

The input attention mechanism determines which recurrent modules activate at each timestep by calculating attention scores based on module hidden states (queries) and incoming environmental data (keys). This selective activation allows only relevant RIMs to process new information, with each unit specializing in different environmental aspects.

Communication attention governs interactions between active RIMs, facilitating efficient information exchange. We modified the original RIM architecture with positional encoding and layer normalization with skip connections during the communication step:

$$h_{t+1,k} = \text{LayerNorm}\left(\tilde{h}_{t,k} + \text{Softmax}\left(\frac{Q_{t,k} \cdot K_{t,:}^T}{\sqrt{d_e}}\right) \cdot V_{t,:}\right) \tag{6}$$

where $Q_{t,k}$, $K_{t,:}$, and $V_{t,:}$ are the query vector from module k, and key and value vectors from all modules. The skip connection preserves gradient flow while normalization prevents any module from dominating the representation.

In each agent's policy network, we remove the RIM's input attention bottleneck since slot attention provides similar cross-attention without blocking gradients based on top-k modules. Following slot attention, each slot representation feeds directly into a separate recurrent module, enabling the network to track temporal changes across specialized features and interpret sequential decision-making of other agents.

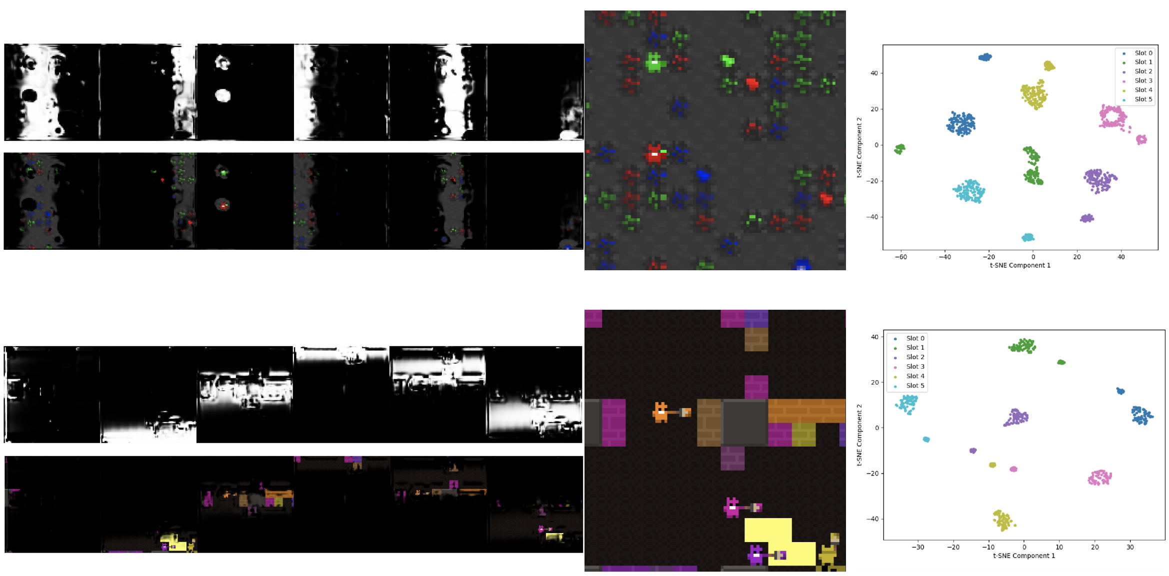

Figure 2: Visual analysis of object-centric representations during pre-training. Left: Attention masks (top) and masked input images with attention mask (bottom). Right: Reconstructed images (left) and compositional consistency with orthogonal projection loss in latent spaces (right) for both Allelopathic Harvest (top) and Territory Room (bottom) environments.

4 Experiments

The Melting Pot suite is a comprehensive multi-agent evaluation environment designed to test the generalization capabilities of MARL algorithms in diverse, complex social scenarios (Agapiou et al., 2023). In our experiments, we focus on three distinct substrates within this suite: Allelopathic Harvest, Territory Rooms, and Prisoner's Dilemma in the Matrix Arena. Allelopathic Harvest involves 16 players on a 29×30 plane populated with 348 berry plants in three colors. Each berry's ripening probability is dynamically modulated by the overall abundance of berries of its color, and agents

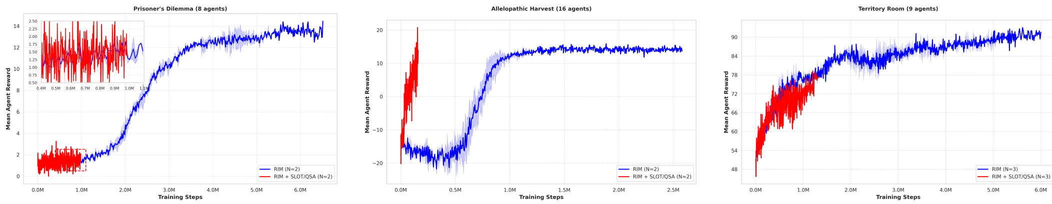

Figure 3: Performance comparison of RIM (blue) versus SLOT ATTENTION/QSA+RIM (red) across three Meltingpot environments. **Prisoner's Dilemma**: Both approaches initially perform similarly. **Allelopathic Harvest**: RIM+SLOT/QSA initially reaches higher rewards (peak ∼20) very quickly, while RIM recovers from initial negative rewards (-20) very slowly to achieve stable positive performance after 1M steps. **Territory Room**: Both approaches initially perform similarly.

receive enhanced rewards when harvesting their preferred berry type. Strategic planting, harvesting, and a zap mechanism that temporarily disables opponents create a competitive yet cooperative dynamic. Territory Rooms is a 9-player environment where each agent aims to claim territory by painting walls with its unique color. Agents are rewarded for maintaining control over their territory, but the risk of elimination through repeated zaps adds a layer of strategic depth. In Prisoner's Dilemma in the Matrix Arena, 8 agents navigate a setting inspired by the classic social dilemma: mutual cooperation yields moderate rewards, unilateral defection yields higher individual payoffs at the expense of the cooperator, and mutual defection results in low rewards. This substrate challenges agents to balance individual gains against collective well-being in a zero-shot generalization context.

These experiments collectively demonstrate that effective multi-agent learning in dynamic social environments requires not only robust perception and decision-making but also the capacity to generalize to novel co-player behaviors and roles. Notably, each substrate tests different facets of cooperative and competitive dynamics: Allelopathic Harvest highlights the tension between immediate consumption and long-term collective benefit through strategic planting, Territory Rooms stresses territorial control and the risk-reward trade-offs inherent in resource acquisition, and Prisoner's Dilemma forces agents to weigh individual incentives against collective outcomes in a classic social dilemma. The Melting Pot protocol (Leibo et al., 2021) provides a scalable framework by pairing physical substrates with unfamiliar background populations, thereby fostering the emergence of socially generalizable behaviors. This combination of diverse substrates and a stringent evaluation protocol underscores the critical role of modular, adaptable architectures in achieving human-like social cognition in MARL—making it an ideal testbed for our architecture.

5 Results

Figure 3 presents the performance comparison between our two primary approaches—standard RIM (blue) and our proposed MAPLE architecture with slot attention (red)—across three distinct Melting Pot environments. Each environment tests different aspects of multi-agent coordination and social dynamics.

In Prisoner's Dilemma (8 agents), both approaches follow similar initial learning trajectories, with MAPLE showing slightly faster early convergence. Through architectural analysis, we determined that GRU units were more effective in this strategic environment, with a 6-module, 5 top-k configuration achieving the best balance between representational capacity and selective attention. The game-theoretic nature of this environment appears to benefit from higher module activation rates, allowing agents to maintain more comprehensive models of opponents' strategies.

For Allelopathic Harvest (16 agents), MAPLE with slot attention demonstrates dramatically faster learning, reaching peak rewards (20) within approximately 200K steps, while standard RIM initially struggles with negative rewards (-20) before gradually recovering. This difference highlights how object-centric representations obtained during pre-training particularly benefit environments with numerous distinct entities (agents and resources). Using LSTM units with 6 modules and 4 top-k activation, MAPLE's slot attention enables quicker identification and specialization toward critical environmental features.

In Territory Room (8 agents), both approaches again show comparable initial performance, with standard RIM eventually achieving higher asymptotic performance. The spatial nature of this environment benefits from LSTM units (6 modules, 4 top-k) which develop specialized spatial representations over extended training. Notably, our analysis of module activation patterns (Figure 4) reveals that maintaining appropriate sparsity (6 total modules with 4 active) yields optimal performance, confirming the importance of selective attention in complex spatial coordination tasks.

Note that computational constraints limited our MAPLE experiments to approximately 1 million training steps, as the attention mechanisms in the slot model significantly increased computational requirements compared to standard RIM model. Despite these constraints, MAPLE demonstrates compelling advantages in learning efficiency, particularly in environments with complex entity interactions. Figure 4 illustrates how performance scales with hidden state dimension across different RIM configurations. Each configuration is denoted as RIM_x_y, where x represents the total number of modules and y indicates how many modules are actively processing information at each timestep (top-k parameter). We observe that while all configurations show improved performance with larger hidden sizes, the relationship between module count and activation sparsity significantly impacts scaling efficiency. Notably, the configuration with 6 total modules and 4 active modules (RIM_6_4) achieves the highest final performance, suggesting that maintaining a balance between capacity (total modules) and sparsity (proportion of inactive modules) leads to more effective resource allocation (Parascandolo et al., 2018). This aligns with our hypothesis that selective activation enables more specialized processing by reducing interference between modules handling different aspects of the environment (Fedus et al., 2022).

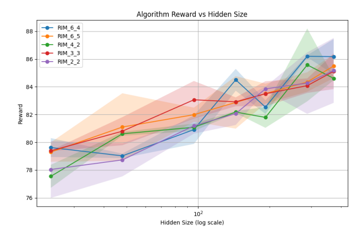

Figure 4: Scaling behavior of different RIM configurations (modules and top-k) with respect to hidden state size in Territory Room substrate.

We study the emergent capability in our agents by computing the mutual information between the joint actions of a pair of players, given the hidden states of one of them, before and after training, as follows:

$$I(a_i; a_j \mid h_{ik}) = \mathbb{E}_{p(a_i,a_j,h_{ik})}\left[\log \frac{p(a_i, a_j \mid h_{ik})}{p(a_i \mid h_{ik})p(a_j \mid h_{ik})}\right]$$

where h_{ik} is the k-th module of agent i and $a_.$ represents actions. Figure 5 illustrates how specific modules within Agent 0 develop specialized representations of other agents' behaviors in Prisoners Dilemma. Module 1 maintains consistently high mutual information with Agents 2, 5, and 7, while Module 2 shows strong connections with Agent 1. This specialization remains stable across episodes (middle) and contributes to the formation of distinct agent clusters based on information processing patterns (right), where agents naturally organize into two groups (0,6,7,1,3 and 2,4,5). Importantly, these specialized representations suggest that modular architectures naturally facilitate social cognition by compartmentalizing agent-specific information processing (Rabinowitz et al., 2018; Jaques et al., 2019; Frith & Frith, 2006).

6 Conclusion

We presented MAPLE, a novel brain-inspired modular architecture that enhances both interpretability and performance in multi-agent reinforcement learning through specialized representation learning. By integrating pre-trained slot attention with recurrent independent mechanisms, MAPLE dis-

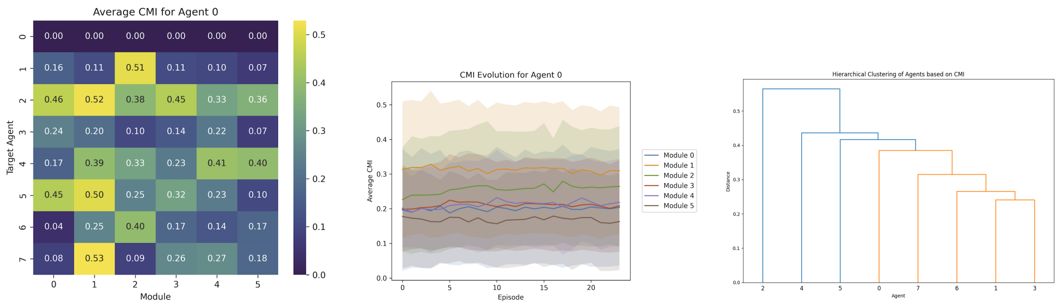

Figure 5: Conditional Mutual Information (CMI) analysis for Agent 0 in Prisoner's Dilemma with RIM and GRU units architecture: Left: Average CMI heatmap by module and target agent, Middle: Module-specific CMI evolution across episodes, Right: Hierarchical clustering of agents based on CMI patterns, revealing two distinct agent clusters (agents 0,1,3,6,7 vs. agents 2,4,5)

entangles complex environmental features into meaningful, object-centric representations that improve coordination and generalization across diverse social settings.

Our approach makes several key contributions to interpretable MARL. The orthogonal latent representations directed through modular recurrent networks enable different modules to specialize in processing distinct environmental features, creating naturally interpretable decision pathways. We employed a selective fine-tuning approach using LoRA that permits efficient adaptation of pre-trained representations while preserving their specialized structure. To address the problem of neural plasticity loss, we implemented weight clipping and shrink-and-perturb methods for our RIM+Slot attention architecture. These techniques systematically identify and revitalize underutilized neurons, maintaining network adaptability throughout the training process and preventing premature convergence to suboptimal representations.

Through evaluation across three Melting Pot environments, we found that MAPLE's modular attention mechanisms particularly excel in complex multi-entity scenarios like Allelopathic Harvest, where they dramatically accelerate learning compared to standard approaches. Our information-theoretic analysis revealed that different modules develop specialized representations of other agents' behaviors, providing quantitative evidence for emergent Theory of Mind capabilities without explicit social modeling objectives.

The specialization of neural modules demonstrated in our work parallels functional specialization observed in human neural systems, suggesting promising directions for developing AI systems that are not only more effective but also more aligned with human cognitive structures. By advancing modular architectures that naturally develop interpretable representations, MAPLE represents a step toward reinforcement learning systems that can learn effectively in complex social environments while remaining transparent and understandable to human observers.

Broader Impact Statement

Our modular approach to MARL has significant implications for AI alignment and safety, as interpretable agent representations enable more reliable verification that systems behave according to human expectations and values. MAPLE's specialized modules allow developers to monitor emergent social behaviors, potentially preventing unintended coordination strategies that might arise in black-box systems. The improved transparency that our approach offers, can help to mitigate risks by making it easier to identify, understand, and correct undesirable emergent behaviors.

Appendix

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Supplementary Materials

The following content was not necessarily subject to peer review.

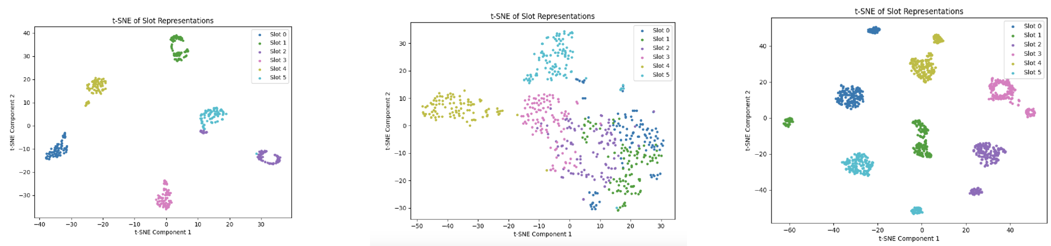

Figure 6: t-SNE visualization of slot representations during pre-training under different regularization strategies. Each color represents a different slot (0-5). These figures demonstrate how our regularization techniques effectively promote specialization and disentanglement in slot representations, which is critical for interpretable multi-agent learning. Left: Slot attention with orthogonal projection regularization. Middle: Slot attention with compositional consistency loss. Right: Slot attention with both regularization methods.

Table 1: Hyperparameter configurations for different environments in MAPLE experiments. The table highlights key differences in model architecture and optimization parameters across environments. Note: Entropy Coefficient values are shown in format 'initial→final', indicating annealing from the higher initial value to the lower final value over the course of training. This gradual reduction encourages exploration early in training while promoting exploitation of learned policies later.

Hyperparameter	**Territory Rooms**	**Prisoner's Dilemma**
Architecture Parameters		
RNN Module	LSTM	GRU
RIM Units	6	6
RIM Top-K	4	5
Hidden Size	300	300
Slot Attention	False	False
Optimization Parameters		
Learning Rate (Actor)	5.0×10^{-5}	9.0×10^{-5}
Learning Rate (Critic)	5.0×10^{-5}	9.5×10^{-5}
Entropy Coefficient	$0.01 \rightarrow 0.005$	$0.1 \rightarrow 0.003$
Value Loss Coefficient	0.75	0.5
PPO Clip Parameter	0.1	0.5
Max Gradient Norm	0.05	0.075
Grad Clip	0.2	Not specified
Gain	0.01	0.05
Huber Delta	Not specified	10.0
Training Parameters		
Number of Agents	9	8
Rollout Threads	15	13
Total Environment Steps	5.5M	60M
Entropy Anneal Duration	600K	800K
Warmup Updates	100K	50K
Cooldown Updates	100K	50K